



IND AS 109 • EXPECTED CREDIT LOSS • TRADE RECEIVABLES

ECL-Pro

An offline, AI-built Expected Credit Loss engine and statutory audit working paper generator for trade receivables under the Ind AS 109 simplified approach.

Presented by **CA ABINASH PARIDA**

AICA Level-2 Capstone Project • Batch B-86 • Built with Google AI Studio

12.64%

Portfolio ECL, base case

6

Ageing buckets in the matrix

A–F

Audit working paper sections

The problem this solves

1

Every audit needs it

Ind AS 109 requires a lifetime ECL allowance on trade receivables for every entity applying Ind AS — not just banks and NBFCs.

2

Excel is the default tool

Provision matrices are built manually each year: ageing pasted from Tally, buckets typed by hand, percentages hard-coded in cells.

3

Errors are hard to detect

Negative balances, duplicate invoices, blank due dates and non-numeric amounts flow straight into the provision undetected.

4

Poor audit trail

No documented methodology, no forward-looking justification, no sign-off trail — a recurring review and NFRA/peer-review comment.

What the project set out to build

COMPUTE

A defensible provision matrix — six ageing buckets, historical default rates, forward-looking overlay, invoice-level ECL.

VALIDATE

Screen the receivables ledger before it is used: negative balances, duplicates, missing due dates, blank customers.

DOCUMENT

Generate a Standards-on-Auditing compliant working paper (A–F) with preparer, reviewer and partner sign-off.

DISCLOSE

Produce the Ind AS 109 note, the journal entry and the Schedule III presentation ready to paste into financials.

EXPORT

Push the whole file — summary, invoice matrix, exceptions, sign-off — into a four-sheet Excel workbook.

RUN OFFLINE

No install, no server, no client data leaving the machine; a single browser build any CA can open.

How ECL-Pro works, end to end

01

INPUT

Excel / CSV receivables ledger uploaded, or rows added manually. Sample and full benchmark datasets ship with the app.

02

VALIDATE

Five rule classes run over every row; errors and warnings are listed as an exceptions report before any number is trusted.

03

AGE & RATE

Days past due computed from due date against the reporting date; each invoice mapped to one of six buckets and its default rate.

04

ADJUST

Historical rate × forward-looking macroeconomic coefficient (Optimistic 0.85 / Base 1.00 / Pessimistic 1.30).

05

OUTPUT

Executive summary, invoice matrix, disclosures, journal entry, audit working paper A-F, Excel and CSV export.

Core formula: $\text{ECL per invoice} = \text{Outstanding Amount} \times (\text{Historical Default Rate for its ageing bucket} \times \text{Forward-looking Multiplier})$

Lifetime ECL, simplified approach — Ind AS 109 paragraph 5.5.15 read with Implementation Guidance B5.5.35. No 12-month / stage-wise split is required for trade receivables without a significant financing component.

The provision matrix and forward-looking overlay

Ageing bucket	Days past due	Historical default %	Optimistic ×0.85	Base ×1.00	Pessimistic ×1.30
Current	On or before due date	0.20%	0.17%	0.20%	0.26%
1–30	1 to 30 days	0.80%	0.68%	0.80%	1.04%
31–60	31 to 60 days	2.10%	1.79%	2.10%	2.73%
61–90	61 to 90 days	4.50%	3.83%	4.50%	5.85%
91–180	91 to 180 days	9.00%	7.65%	9.00%	11.70%
Above 180	More than 180 days	22.00%	18.70%	22.00%	28.60%

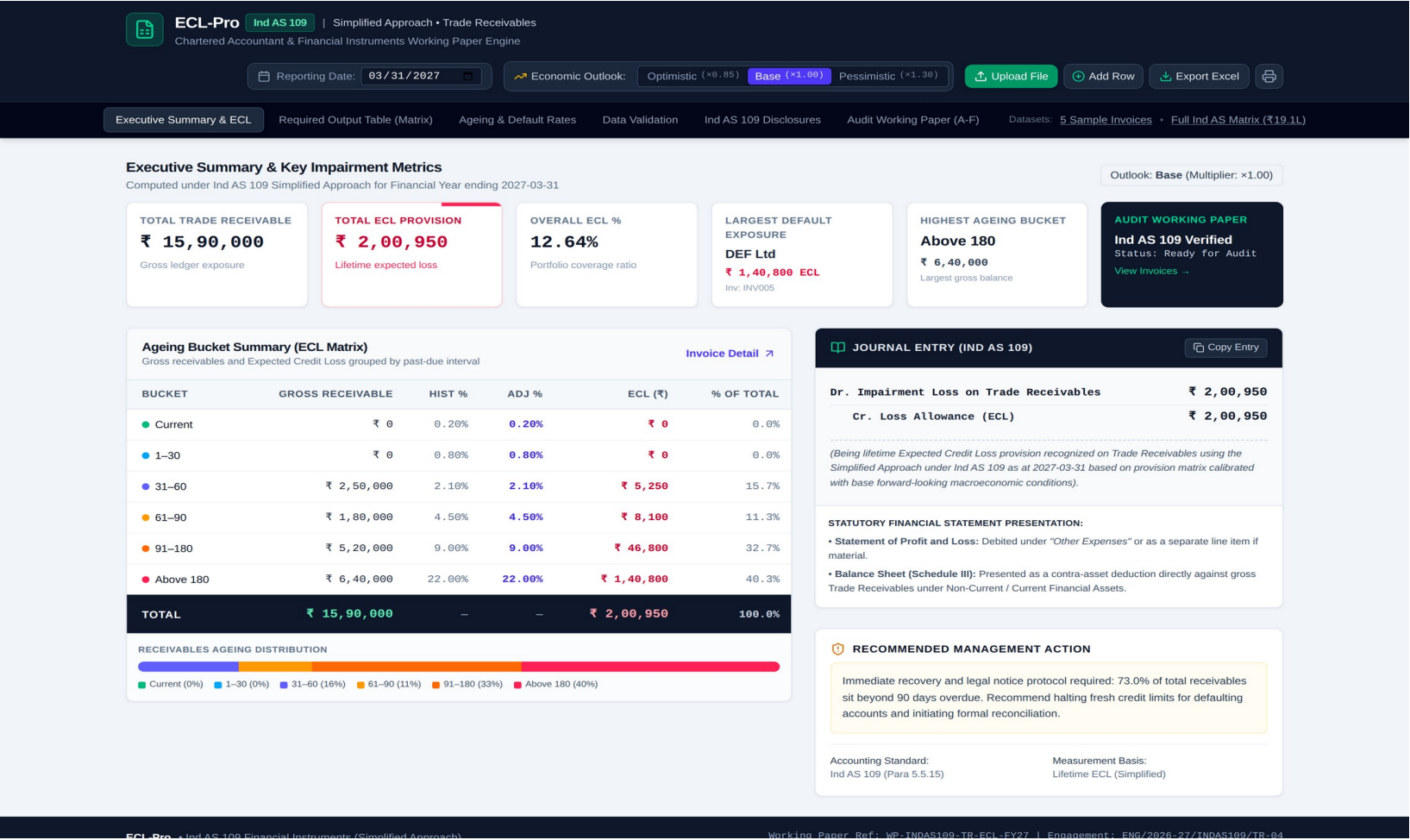
Bucket assignment

Days past due = Reporting date – Due date. Where the ledger already carries a days-outstanding figure, that value is used and the row is flagged in the exceptions report.

Forward-looking overlay

A single macroeconomic coefficient scales every historical rate, so the judgement is explicit, documented in one place and testable by the reviewer — instead of buried in cell formulas.

Executive summary and key impairment metrics



What the reviewer sees first

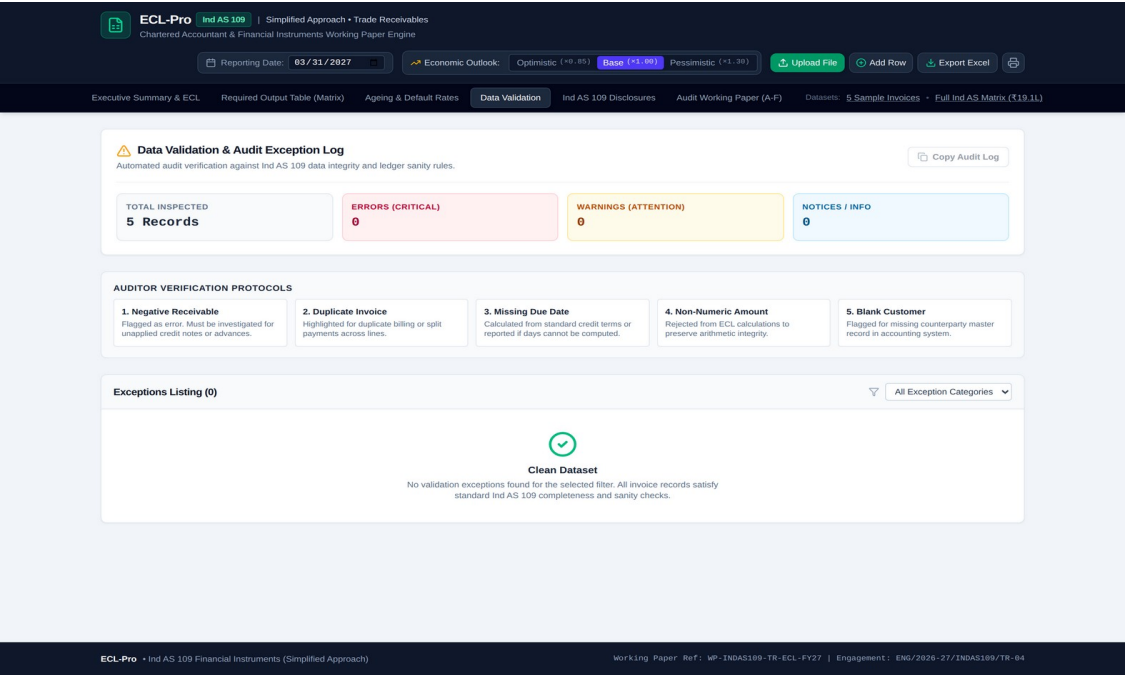
- Six headline metrics: gross receivable, ECL provision, coverage %, largest single exposure, heaviest bucket, audit status.
- The full ageing bucket matrix with historical and adjusted rates side by side.
- The Ind AS 109 journal entry, ready to copy.
- A generated management action note driven by the >90-day concentration.

SA 230 / SA 540 documentation

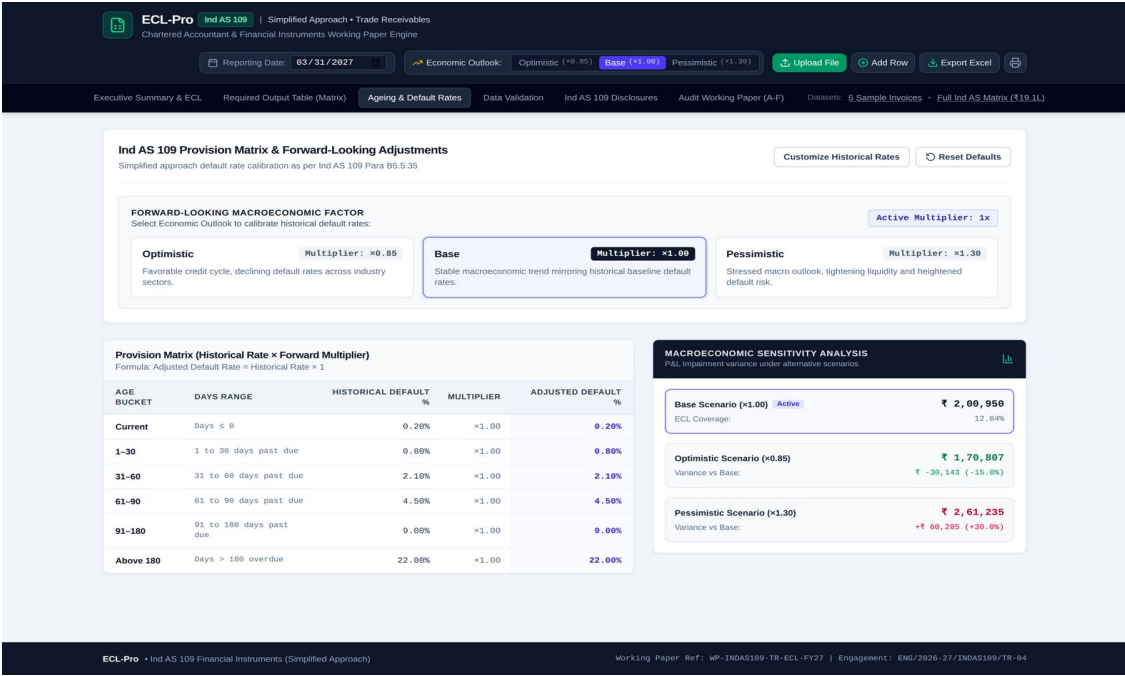
- A Objective of the procedure
- B Source of data and reconciliation
- C Methodology applied
- D Computation and results
- E Audit conclusion
- F Preparer, reviewer and partner sign-off

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Data validation and rate calibration



Exceptions report — five rule classes run before any provision is computed.



Rate matrix — historical percentages are editable and the outlook multiplier is applied live.

Validation rules: negative receivable • duplicate invoice number • missing due date • non-numeric amount • blank customer name

Illustrative run — five-invoice sample portfolio

TOTAL TRADE RECEIVABLE

₹ 15,90,000

Gross ledger exposure

TOTAL ECL PROVISION

₹ 2,00,950

Lifetime expected loss

OVERALL ECL %

12.64%

Portfolio coverage ratio

CONCENTRATION >90 DPD

73.0%

Of gross receivables

Ageing bucket	Gross receivable	Adj. rate	ECL	% of gross
31-60	₹ 2,50,000	2.10%	₹ 5,250	15.7%
61-90	₹ 1,80,000	4.50%	₹ 8,100	11.3%
91-180	₹ 5,20,000	9.00%	₹ 46,800	32.7%
Above 180	₹ 6,40,000	22.00%	₹ 1,40,800	40.3%
TOTAL	₹ 15,90,000	—	₹ 2,00,950	100.0%

Sensitivity to the outlook		
Optimistic ×0.85	₹ 1,70,807	10.74%
Base ×1.00	₹ 2,00,950	12.64%
Pessimistic ×1.30	₹ 2,61,235	16.43%

Recommended action generated by the model: 73.0% of receivables sit beyond 90 days — halt fresh credit limits for defaulting accounts and initiate formal reconciliation and legal notice protocol.

What comes out of a run

Excel workbook



Four sheets: ECL_Summary, Invoice_ECL_Matrix, Validation_Exceptions, Audit_WP_Signoff — the complete file in one download.

CSV matrix



The required output table: customer, outstanding, days, bucket, historical %, adjusted %, ECL.

Ind AS 109 note



Disclosure paragraph, loss allowance movement and Schedule III presentation of the contra-asset.

Journal entry



Dr Impairment Loss on Trade Receivables / Cr Loss Allowance, with narration and reporting date.

Audit working paper



Sections A-F with reference, engagement code, status and three-level sign-off, printable from the browser.

Exceptions report



Every flagged row with severity, rule breached and the original data for follow-up with the client.

How the application was actually built

01

Prompt

The Ind AS 109 requirement — buckets, rates, validation rules, required output table — was written as a domain specification, not as code instructions.

02

Generate

Google AI Studio produced the React + TypeScript application: calculation engine, six views, Excel export.

03

Verify

Every computed figure was tied back by hand to an independently prepared provision matrix before the build was accepted.

04

Iterate

Refinement prompts added the audit working paper, the exceptions report, the outlook multiplier and the disclosure view.

Technology stack

React 19 + TypeScript

Vite build

Tailwind CSS

SheetJS (xlsx) export

Google AI Studio

Runs fully client-side

Limitations and where it goes next

Limitations today

- Default rates are entered by the user, not derived from the client's own roll-rate history.
- A single macroeconomic multiplier stands in for a full probability-weighted scenario model.
- Simplified approach only — no three-stage model for loans or debt instruments.
- No direct Tally / ERP link; data arrives by Excel or CSV upload.
- State lives in the browser session — nothing is persisted between runs.

Planned enhancements

- Derive historical default rates automatically from three to five years of uploaded ageing data.
- Multi-scenario probability weighting with documented scenario weights.
- Direct Tally / Zoho / ERP receivables import.
- Customer-segment and geography-wise matrices within one engagement.
- PDF working paper export with digital sign-off and version history.

The model is a working-paper aid. Rate selection, outlook and the final provision remain the professional judgement of the engagement partner.

Demonstration sequence

- 1 Open the build in the browser — no install, no login, no server call.
- 2 Load the five-invoice sample dataset; set the reporting date to 31-03-2027.
- 3 Open Data Validation and show the exceptions raised on the ledger.
- 4 Move to Ageing & Default Rates; edit a historical rate and switch the outlook to Pessimistic to show the provision move live.
- 5 Return to the Executive Summary; read the ECL, the bucket matrix and the journal entry.
- 6 Open the Audit Working Paper (A-F) and the Ind AS 109 disclosure note.
- 7 Export the Excel workbook and open the four sheets.

Run time

6–8

minutes, end to end

Against roughly half a day to build the same provision matrix by hand in Excel for a comparable ledger — and without the ageing, validation and documentation being re-keyed.

THANK YOU

Questions and review comments welcome.

ECL-Pro • Expected Credit Loss working paper engine for trade receivables under Ind AS 109 (Simplified Approach)

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